SCHOOL OF ENGINEERING AND TECHNOLOGY

COURSEWORK FOR

BSC (HONS) IN COMPUTER SCIENCE
BSC (HONS) INFORMATION TECHNOLOGY
BACHELOR OF SOFTWARE ENGINEERING (HONS)
BSC (HONS) INFORMATION SYSTEMS (DATA ANALYTICS)
BSC (HONS) INFORMATION TECHNOLOGY (COMPUTER NETWORKING AND SECURITY)

YEAR 1; ACADEMIC SESSION APRIL 2024

SEG1201: DATABASE FUNDAMENTALS

Parts 1 – 4 Due dates: Week 13 Monday, 9am

Part 5 – refer to the given schedule

SEG1201: DATABASE FUNDAMENTALS

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INSTRUCTIONS

- This final assessment contributes 50% to your final grade.
- This five-member group assignment (or minimum 3-member group) is primarily for Course Learning Outcome 2 - Implement a database design group project using appropriate tools such as Oracle SQL.

IMPORTANT

The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

Academic Honesty Acknowledgement

"We LIANG YRENN DA, OOI JIN YON, YASHINI A/P CHANDRAN, MUHAMMAD IKRAM BIN MOHAMED verify that this paper contains entirely our own work. We have not consulted with any outside person or materials other than what was specified (an interviewee, for example) in the assignment or the syllabus requirements. Further, we have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. We realize the penalties *(refer to page 16, 5.5, Appendix 2, page 44 of the student handbook diploma and undergraduate programme)* for any kind of copying or collaboration on any assignment."
LIANG YRENN DA (2024/7/18), OOI JIN YON (2024/7/18), YASHINI A/P CHANDRAN (2024/7/18), MUHAMMAD IKRAM BIN MOHAMED (2024/7/18)

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Part 1: Generate a case scenario & draw an ERD (10 marks)

a) Travoka Travel Agency Scenario

Travoka is a comprehensive travel agency platform designed to help customers plan and book their travel arrangements seamlessly. The platform efficiently manages user registrations, travel package listings, bookings, payments, reviews, destinations, invoices, accommodations, and optional elements like promotions and transportation. In Travoka's system, each travel agent works for a specific branch, and a branch must employ at least five agents to ensure adequate service coverage. An agent can serve up to four customers, and each customer is assigned to only one agent, fostering personalized service and better customer relationships. The salary will be updated when the agent receive their salary monthly.

Customers can make up to two reservations through Travoka, with each reservation linked to a single customer to maintain clarity and organization in booking records. A customer can write a review for a reservation, but only after completing their travel, ensuring that reviews are based on complete experiences. Every reservation corresponds to one travel package, simplifying the management of travel plans. To streamline financial transactions, each reservation must be associated with a payment, ensuring that all bookings are secured financially.

Travoka's packages can include multiple destinations, allowing for diverse and customized travel experiences. Conversely, a destination can be part of multiple packages, providing flexibility in itinerary planning. Each branch offers multiple travel packages, and any given package can be available through various branches, broadening customer access to diverse travel options. A destination can only be involved in one package multiple time, and a package can feature two destinations, providing balanced and varied travel experiences.

Travoka ensures that each payment is linked to one reservation, but a reservation can accommodate multiple payments. This flexibility in payment management caters to customers' financial preferences and needs. Additionally, each reservation generates an invoice, detailing the services and charges associated with the booking, which helps maintain clear financial records. Travel packages can also include optional elements such as promotions and transportation, enhancing the travel experience.

Each entity in Travoka has specific attributes for their database:

- **Branch** (BranchID, BranchName, BranchContact, BranchAddress, BranchEmail, BranchManager)
- **Agent** (AgentID, BranchID, AgentName, AgentPhone, AgentEmail, DateHired, DateRetire, AgentCommisionRate, AgentAddress, SupervisorID)
- **Customer** (CustomerID, Cusname, CusContact, CusAddress, CusEmail, AgentID)
- **Reservation** (ReservationID, CustomerID, AgentID, ReserveDate, PackageID, Status, ReviewID)
- **Review** (ReviewID, CustomerID, ReservationID, ReviewDesc, ReviewDate, Rating)
- **Package** (PackageID, PackageName, PackageDesc, PackPrice, Duration, TravelStartDate. TravelEndDate)
- **Payment** (PaymentID, ReservationID, PaymentType, PaymentDate, Amount)
- **Invoice** (InvoiceID, PaymentID, InvoiceDate)
- **Destination** (DestinationID, DestinationName, DestinationDesc, Distance, DestinationType, PackageID)

- **Accommodation** (AccomodationID, AccomodationType, AccomadationName, AccomadationAddress, PackageID)
- **Promotion** (PromotionID, PromotionType, PromotionName, PromotionStartDate, PromotionEndDate, DiscountPackageID, TermsAndConditions)
- **Transportation** (TransportID, TransportType, TransportCompany, Departure, Arrival, PackageID)
- **Salary** (SalaryID, AgentID, BaseSalary, Commission, TotalSalary, PayDate, Bonus)

By integrating these entities and their attributes, Travoka ensures a smooth, organized, and efficient travel planning and booking experience for all users.

b) ERD diagram

c) Business rules

1) An agent works for one branch, and a branch must have at least 5 agents.
2) An agent can serve up to 4 customers, and a customer can only be served by one agent.
3) A customer can make up to 2 reservations, but each reservation is made by one customer.
4) A reservation can contain 0 or 1 review for a reservation only after completing the travel.
5) A customer can write 0 or many reviews.
6) Each reservation is for one package.
7) Each reservation must have an associated payment.
8) A reservation can have 2 packages, and each package can be part of multiple reservation concurrently.
9) A branch offers multiple packages, and each package can be offered by multiple branches.
10) A package can have up to 2 destinations, and each destination can be part of multiple packages concurrently.
11) Each payment must be linked to one reservation, and a reservation can have one payment only.
12) Each payment generates an invoice, and each invoice is linked to one payment.
13) A package includes at least one accommodation, and an accommodation is linked to one package.
14) A package can have optional promotions, and each promotion is linked to one package.
15) A package can have optional transportation, and each transportation option is linked to one package.
16) One agent can have zero or one supervisor, but a supervisor must have 3 subordinates (From agent)
17) An agent's commission rate for salary must be greater than 0 and should not exceed 0.1 (10%).

Part 2: Design a relational data (35 marks)

a) Relational Database Model (15 marks)

Identify the primary & an alternate key, and foreign (if any) keys for each table. List the attributes for each entity.

Table: Branch *BranchID* *BranchName* *BranchContact* *BranchAddress* *BranchEmail* *BranchManager* *Alternate key: BranchContact* *FOREIGN KEY **BranchManager** REFERENCES Table **Agent***	***Table: Agent*** *AgentID* *BranchID* *AgentName* *AgentPhone* *AgentEmail* *AgentAddress* *DateHired* *DateRetire* *AgentCommisionRate* *SupervisorID* *Alternative Key: AgentPhone* *FOREIGN KEY **BranchID** REFERENCES Table **Branch*** *FOREIGN KEY **SupervisorID** REFERENCES Table **Agent (Unary)***
Table: Customer *CustomerID* *Cusname* *CusContact* *CusAddress* *CusEmail* *AgentID* *Alternative Key: CusContact* *FOREIGN KEY **AgentID** REFERENCES TABLE **Agent***	***Table: Reservation*** *ReservationID* *CustomerID* *AgentID* *ReserveDate* *TravelStartDate* *TravelEndDate* *Status* *ReviewID* *Alternative Key: CustomerID + ReserveDate + TravelStartDate* *FOREIGN KEY **CustomerID** REFERENCES Table **Customer***
Table: Review	***Table: Package_Reservation***

ReviewID *CustomerID* *ReservationID* *ReviewDesc* *ReviewDate* *Rating* *Alternative Key: CustomerID + ReservationID* *FOREIGN KEY **CustomerID** REFERENCES Table **Customer*** *FOREIGN KEY **ReservationID** REFERENCES Table **Reservation***	*PackageID* *ReservationID* *Quantity* *Price* *Alternative Key: None* *FOREIGN KEY **PackageID** REFERENCES Table **Package*** *FOREIGN KEY **ReservationID** REFERENCES Table **Reservation***
Table: Package *PackageID* *PackageName* *PackageDesc* *PackPrice* *TravelStartDate* *TravelEndDate* *Alternative Key: PackageName + PackPrice + TravelStartDate*	***Table: Payment*** *PaymentID* *ReservationID* *PaymentType* *PaymentDate* *Amount* *PaymentStatus* *Alternative Key: ReservationID + PaymentDate*
Table : Invoice *InvoiceID* *PaymentID* *InvoiceDate* *Alternative Key: PaymentID* *FOREIGN KEY **PaymentID** REFERENCES **Payment***	***Table Destination*** *DestinationID* *DestinationName* *DestinationDesc* *Distance* *DestinationType* *PackageID* *Alternative Key: PackageID + destinationName* *FOREIGN KEY **PackageID** REFERENCES **Package***
Table : Promotion *PromotionID* *PromotionType* *PromotionName* *PromotionStartDate* *PromotionEndDate*	***Table : Transportation*** *TransportID* *TransportType* *TransportCompany* *Departure* *Arrival*

TermsAndConditions *Discount* *PackageID* *Alternative Key: PackageID + PromotionName + PromotionStartDate* *FOREIGN KEY* ***PackageID*** *REFERENCES* ***Package***	*PackageID* *Alternative Key: TransportType + Departure + PackageID* *FOREIGN KEY* ***PackageID*** *REFERENCES* ***Package***
Table : Accommodation *AccommodationID* *AccommodationType* *AccommadationName* *AccommadationAddress* *PackageID* *Alternative Key: PackageID + AccomodationName* *FOREIGN KEY* ***PackageID*** *REFERENCES* ***Package***	***Table : Package_Destination*** *PackageID* *DestinationID* *Quantity* *Price* *Alternative Key: NONE* *FOREIGN KEY* ***PackageID*** *REFERENCES Table* ***Package*** *FOREIGN KEY* ***DestinationID*** *REFERENCES Table* ***Destination***
Table : Salary SalaryID AgentID BaseSalary Commission TotalSalary PayDate Bonus *FOREIGN KEY* ***AgentID*** *REFERENCES* ***Agent***	

b) Constraints (10 marks)

No	Constraint Type/Description	Justification
1	PackPrice CHECK Constraint	Packages should have a price greater than zero. This ensures that no package is entered with an invalid or free price, which could lead to revenue loss or data inconsistency.
2	ReviewRating CHECK Constraint	Reviews should have a rating between 1 and 5. This ensures that all ratings fall within a predefined range, making the rating data consistent and meaningful.
3	ReserveDate CHECK Constraint	Reservations should not be made for dates in the past. This ensures that all reservation dates are valid and makes logical sense for future bookings.
4	PaymentType CHECK Constraint	The payment type should be limited to specific values (e.g., 'Credit Card', 'Debit Card', 'Cash', 'Bank Transfer'). This ensures consistency in the types of payments accepted and helps in processing payments correctly.
5	Customer Contact Number Check Constraint	Ensuring that the customer's contact number follows a valid Malaysian phone number format helps maintain data accuracy and ensures that all contact numbers are consistent and usable for communication

c) Indexes (10 marks)

No	Index Type/Description	Justification
1	DateHired Index on Agent Table	The HR department often needs to retrieve agents based on their hire dates for reporting purposes, such as determining the length of service, eligibility for benefits, and anniversary tracking. Indexing this field will significantly speed up these queries.
2	ReserveDate Index on Reservation Table	Reservation queries are frequently performed based on the reservation date to manage and organize upcoming trips. Indexing this field will make it faster to retrieve reservations for a particular date or date range.
3	CustomerID Index on Review Table	Often, customer service representatives need to pull up all reviews written by a particular customer quickly to address feedback or complaints. Indexing CustomerID in the Review table will expedite this process.
4	DestinationName Index on Destinations Table	The travel planning team frequently searches for destinations by name to create or update travel packages. Indexing DestinationName will expedite these search operations.
5	InvoiceDate Index on Invoice Table	The finance department often generates reports and tracks invoices based on the invoice date. Indexing InvoiceDate will improve the performance of these queries.

Part 3: Implement a database (15 marks)

```
-----------------------------------------------------------------------------------------
-- Travoka Travel Agency Database
-----------------------------------------------------------------------------------------

-- The Travoka Travel Agency database consists of the following tables:
-- Branch - stores the details of each branch.
-- Agent - stores the particulars of travel agents.
-- Customer - stores the particulars of customers such as name, contact, etc.
-- Package - stores the details of travel packages.
-- Reservation - stores the details of customer reservations.
-- Review - stores reviews given by customers after their travel.
-- Payment - stores payment details for reservations.
-- Invoice - stores invoice details for payments.
-- Destination - stores details of travel destinations.
-- Promotion - stores promotion details for packages.
-- Transportation - stores transportation details for packages.
-- Accommodation - stores accommodation details for packages.
-- Package_Reservation - links packages and reservations.
-- Package_Destination - links packages and destinations.
-- Salary - stores the particulars of agents salary

-----------------------------------------------------------------------------------------
-- Drop tables if they exist
-----------------------------------------------------------------------------------------
ALTER SESSION SET NLS_DATE_FORMAT='DD-MON-YYYY';
BEGIN
   EXECUTE IMMEDIATE 'DROP TABLE Package_Destination CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Destination CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Promotion CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Transportation CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Accommodation CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Invoice CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Payment CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Package_Reservation CASCADE
CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Review CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Reservation CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Customer CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Agent CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Salary CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Branch CASCADE CONSTRAINTS';
   EXECUTE IMMEDIATE 'DROP TABLE Package CASCADE CONSTRAINTS';
EXCEPTION
   WHEN OTHERS THEN NULL;
```

```
END;
/

-------------------------------------------------------------------------------------
-- Create Branch table
-------------------------------------------------------------------------------------
CREATE TABLE Branch (
    BranchID NUMBER PRIMARY KEY,
    BranchName VARCHAR2(50) NOT NULL,
    BranchContact VARCHAR2(15),
    BranchAddress VARCHAR2(100),
    BranchEmail VARCHAR2(50),
    BranchManager NUMBER
);

-------------------------------------------------------------------------------------
-- Create Agent table
-------------------------------------------------------------------------------------
CREATE TABLE Agent (
    AgentID NUMBER PRIMARY KEY,
    BranchID NUMBER NOT NULL,
    AgentName VARCHAR2(50) NOT NULL,
    AgentPhone VARCHAR2(15),
    AgentEmail VARCHAR2(50),
    AgentAddress VARCHAR2(100),
    DateHired DATE,
    DateRetire DATE,
    AgentCommisionRate NUMBER CONSTRAINT chk_AgentCommisionRate CHECK
(AgentCommisionRate > 0),
    SupervisorID NUMBER,
    FOREIGN KEY (BranchID) REFERENCES Branch(BranchID),
    FOREIGN KEY (SupervisorID) REFERENCES Agent(AgentID)
);

-------------------------------------------------------------------------------------
-- Create Salary table
-------------------------------------------------------------------------------------
CREATE TABLE Salary (
    SalaryID NUMBER PRIMARY KEY,
    AgentID NUMBER NOT NULL,
    BaseSalary NUMBER CONSTRAINT chk_BaseSalary CHECK (BaseSalary > 0),
    Commission NUMBER CONSTRAINT chk_Commission CHECK (Commission >= 0),
    TotalSalary NUMBER CONSTRAINT chk_TotalSalary CHECK (TotalSalary > 0),
    PayDate DATE,
    Bonus NUMBER,
```

```
    FOREIGN KEY (AgentID) REFERENCES Agent(AgentID)
);



-------------------------------------------------------------------------------------
-- Create Customer table
-------------------------------------------------------------------------------------
CREATE TABLE Customer (
    CustomerID NUMBER PRIMARY KEY,
    CusName VARCHAR2(50) NOT NULL,
    CusContact VARCHAR2(15) CONSTRAINT chk_CusContact CHECK (
        REGEXP_LIKE(CusContact, '^(01[0-9]{1}-[0-9]{7,8})|(0[3-9][0-9]-[0-9]{7,8})$')
    ),
    CusAddress VARCHAR2(100),
    CusEmail VARCHAR2(50),
    AgentID NUMBER,
    FOREIGN KEY (AgentID) REFERENCES Agent(AgentID)
);

-------------------------------------------------------------------------------------
-- Create Package table
-------------------------------------------------------------------------------------
CREATE TABLE Package (
    PackageID NUMBER PRIMARY KEY,
    PackageName VARCHAR2(50) NOT NULL,
    PackageDesc VARCHAR2(255),
    PackPrice NUMBER CONSTRAINT chk_PackPrice CHECK (PackPrice > 0),
    TravelStartDate DATE,
    TravelEndDate DATE
);

-------------------------------------------------------------------------------------
-- Create Reservation table
-------------------------------------------------------------------------------------
CREATE TABLE Reservation (
    ReservationID NUMBER PRIMARY KEY,
    CustomerID NUMBER NOT NULL,
    AgentID NUMBER NOT NULL,
    ReserveDate DATE,
    TravelStartDate DATE,
    TravelEndDate DATE,
    Status VARCHAR2(20) CONSTRAINT chk_Status CHECK (Status IN ('Confirmed',
'Pending', 'Cancelled')),
    ReviewID NUMBER,
```

```
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (AgentID) REFERENCES Agent(AgentID)
);

-------------------------------------------------------------------------------
-- Create Review table
-------------------------------------------------------------------------------
CREATE TABLE Review (
    ReviewID NUMBER PRIMARY KEY,
    CustomerID NUMBER NOT NULL,
    ReservationID NUMBER NOT NULL,
    ReviewDesc VARCHAR2(255),
    ReviewDate DATE,
    Rating NUMBER CONSTRAINT chk_Rating CHECK (Rating BETWEEN 1 AND 5),
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (ReservationID) REFERENCES Reservation(ReservationID)
);

-------------------------------------------------------------------------------
-- Create Payment table
-------------------------------------------------------------------------------
CREATE TABLE Payment (
    PaymentID NUMBER PRIMARY KEY,
    ReservationID NUMBER NOT NULL,
    PaymentType VARCHAR2(20) CONSTRAINT chk_PaymentType CHECK (PaymentType
IN ('Credit Card', 'Debit Card', 'Cash', 'Bank Transfer')),
    PaymentDate DATE,
    Amount NUMBER,
    PaymentStatus VARCHAR2(20),
    FOREIGN KEY (ReservationID) REFERENCES Reservation(ReservationID)
);

-------------------------------------------------------------------------------
-- Create Invoice table
-------------------------------------------------------------------------------
CREATE TABLE Invoice (
    InvoiceID NUMBER PRIMARY KEY,
    PaymentID NUMBER NOT NULL,
    InvoiceDate DATE,
    FOREIGN KEY (PaymentID) REFERENCES Payment(PaymentID)
);

-------------------------------------------------------------------------------
-- Create Destination table
-------------------------------------------------------------------------------
```

```
CREATE TABLE Destination (
    DestinationID NUMBER PRIMARY KEY,
    DestinationName VARCHAR2(50),
    DestinationDesc VARCHAR2(255),
    Distance NUMBER,
    DestinationType VARCHAR2(50),
    PackageID NUMBER,
    FOREIGN KEY (PackageID) REFERENCES Package(PackageID)
);

-------------------------------------------------------------------------------
-- Create Promotion table
-------------------------------------------------------------------------------
CREATE TABLE Promotion (
    PromotionID NUMBER PRIMARY KEY,
    PromotionType VARCHAR2(50),
    PromotionName VARCHAR2(50),
    PromotionStartDate DATE,
    PromotionEndDate DATE,
    TermsAndConditions VARCHAR2(255),
    Discount NUMBER,
    PackageID NUMBER,
    FOREIGN KEY (PackageID) REFERENCES Package(PackageID)
);

-------------------------------------------------------------------------------
-- Create Transportation table
-------------------------------------------------------------------------------
CREATE TABLE Transportation (
    TransportID NUMBER PRIMARY KEY,
    TransportType VARCHAR2(50),
    TransportCompany VARCHAR2(50),
    Departure DATE,
    Arrival DATE,
    PackageID NUMBER,
    FOREIGN KEY (PackageID) REFERENCES Package(PackageID)
);

-------------------------------------------------------------------------------
-- Create Accommodation table
-------------------------------------------------------------------------------
CREATE TABLE Accommodation (
    AccommodationID NUMBER PRIMARY KEY,
    AccommodationType VARCHAR2(50),
    AccommodationName VARCHAR2(50),
```

```
    AccommodationAddress VARCHAR2(100),
    PackageID NUMBER,
    FOREIGN KEY (PackageID) REFERENCES Package(PackageID)
);

-------------------------------------------------------------------------------
-- Create Package_Reservation table
-------------------------------------------------------------------------------
CREATE TABLE Package_Reservation (
    PackageID NUMBER,
    ReservationID NUMBER,
    Quantity NUMBER,
    Price NUMBER,
    PRIMARY KEY (PackageID, ReservationID),
    FOREIGN KEY (PackageID) REFERENCES Package(PackageID),
    FOREIGN KEY (ReservationID) REFERENCES Reservation(ReservationID)
);

-------------------------------------------------------------------------------
-- Create Package_Destination table
-------------------------------------------------------------------------------
CREATE TABLE Package_Destination (
    PackageID NUMBER,
    DestinationID NUMBER,
    Quantity NUMBER,
    Price NUMBER,
    PRIMARY KEY (PackageID, DestinationID),
    FOREIGN KEY (PackageID) REFERENCES Package(PackageID),
    FOREIGN KEY (DestinationID) REFERENCES Destination(DestinationID)
);

-------------------------------------------------------------------------------
-- Indexes
-------------------------------------------------------------------------------

-- The HR department often needs to retrieve agents based on their hire dates for reporting purposes, such as determining the length of service, eligibility for benefits, and anniversary tracking--
CREATE INDEX idx_date_hired ON Agent(DateHired);
--Reservation queries are frequently performed based on the reservation date to manage and organize upcoming trips. Indexing this field will make it faster to retrieve reservations for a particular date or date range.--
CREATE INDEX idx_reserve_date ON Reservation(ReserveDate);
--customer service representatives need to pull up all reviews written by a particular customer quickly to address feedback or complaints--
```

```
CREATE INDEX idx_customer_id ON Review(CustomerID);
--travel planning team frequently searches for destinations by name to create or update travel packages. Indexing DestinationName will expedite these search operations.--
CREATE INDEX idx_destination_name ON Destination(DestinationName);
--finance department often generates reports and tracks invoices based on the invoice date.--
CREATE INDEX idx_invoice_date ON Invoice(InvoiceDate);

--------------------------------------------------------------------------------
-- Insert data into Branch table
--------------------------------------------------------------------------------
INSERT INTO Branch VALUES (1001, 'Kuala Lumpur Central', '0321234567', '123 Jalan Bukit Bintang', 'klcentral@travoka.my', NULL);
INSERT INTO Branch VALUES (1002, 'Penang North', '0421234567', '456 Jalan Gurney', 'penangnorth@travoka.my', NULL);
INSERT INTO Branch VALUES (1003, 'Johor Bahru South', '0721234567', '789 Jalan Tebrau', 'johorbahru@travoka.my', NULL);
INSERT INTO Branch VALUES (1004, 'Kota Kinabalu East', '0881234567', '321 Jalan Tanjung Aru', 'kotakinabalu@travoka.my', NULL);
INSERT INTO Branch VALUES (1005, 'Kuching West', '0821234567', '654 Jalan Pending', 'kuching@travoka.my', NULL);

--------------------------------------------------------------------------------
-- Insert data into Agent table
--------------------------------------------------------------------------------
INSERT INTO Agent VALUES (2001, 1001, 'Ahmad Ali', '0123456789', 'ahmad@travoka.my', '10 Jalan Pudu', TO_DATE('2018-01-01', 'YYYY-MM-DD'), NULL, 0.05, NULL);
INSERT INTO Agent VALUES (2002, 1001, 'Siti Aminah', '0198765432', 'siti@travoka.my', '20 Jalan Imbi', TO_DATE('2019-01-01', 'YYYY-MM-DD'), NULL, 0.04, 2001);
INSERT INTO Agent VALUES (2003, 1001, 'Tan Wei Ming', '0112233445', 'tan@travoka.my', '30 Jalan Alor', TO_DATE('2020-02-01', 'YYYY-MM-DD'), NULL, 0.06, 2001);
INSERT INTO Agent VALUES (2004, 1001, 'Lim Pei Ling', '0199988776', 'lim@travoka.my', '40 Jalan Burma', TO_DATE('2017-03-15', 'YYYY-MM-DD'), NULL, 0.05, 2001);
INSERT INTO Agent VALUES (2005, 1002, 'Cheong Kai Lun', '0187766554', 'cheong@travoka.my', '50 Jalan Macalister', TO_DATE('2016-04-20', 'YYYY-MM-DD'), NULL, 0.04, 2004);
INSERT INTO Agent VALUES (2006, 1002, 'Teoh Hock Seng', '0134455667', 'teoh@travoka.my', '60 Jalan Anson', TO_DATE('2015-05-25', 'YYYY-MM-DD'), NULL, 0.06, 2004);
INSERT INTO Agent VALUES (2007, 1003, 'Ravi Subramaniam', '0172233445', 'ravi@travoka.my', '70 Jalan Trus', TO_DATE('2019-06-30', 'YYYY-MM-DD'), NULL, 0.05, NULL);
INSERT INTO Agent VALUES (2008, 1003, 'Suresh Kumar', '0165544332', 'suresh@travoka.my', '80 Jalan Wong Ah Fook', TO_DATE('2018-07-10', 'YYYY-MM-DD'), NULL, 0.04, 2007);
```

```
INSERT INTO Agent VALUES (2009, 1003, 'Tanaka Hiroshi', '0156655443',
'tanaka@travoka.my', '90 Jalan Ibrahim', TO_DATE('2020-08-15', 'YYYY-MM-DD'), NULL,
0.06, 2007);
INSERT INTO Agent VALUES (2010, 1004, 'Rosli Zainal', '0132233456', 'rosli@travoka.my',
'100 Jalan Tun Fuad', TO_DATE('2017-09-20', 'YYYY-MM-DD'), NULL, 0.05, NULL);
INSERT INTO Agent VALUES (2011, 1004, 'Mariam Rahman', '0195544332',
'mariam@travoka.my', '110 Jalan Pantai', TO_DATE('2016-10-25', 'YYYY-MM-DD'), NULL,
0.04, 2010);
INSERT INTO Agent VALUES (2012, 1004, 'Jonathan Ng', '0186655443',
'jonathan@travoka.my', '120 Jalan Lintas', TO_DATE('2015-11-30', 'YYYY-MM-DD'), NULL,
0.06, 2010);
INSERT INTO Agent VALUES (2013, 1005, 'Josephine Tan', '0173344556',
'josephine@travoka.my', '130 Jalan Padungan', TO_DATE('2019-12-05', 'YYYY-MM-DD'),
NULL, 0.05, NULL);
INSERT INTO Agent VALUES (2014, 1005, 'Mohd Arif', '0194455667', 'arif@travoka.my',
'140 Jalan Pending', TO_DATE('2018-01-10', 'YYYY-MM-DD'), NULL, 0.04, 2013);
INSERT INTO Agent VALUES (2015, 1005, 'Christina Wong', '0135566778',
'christina@travoka.my', '150 Jalan Tabuan', TO_DATE('2017-02-15', 'YYYY-MM-DD'), NULL,
0.06, 2013);

-----------------------------------------------------------------------------------
-- Insert data into Salary table
-----------------------------------------------------------------------------------
INSERT INTO Salary VALUES (1, 2001, 5000, 5000 * 0.05, 5000 + (5000 * 0.05) + 500,
TO_DATE('2024-07-31', 'YYYY-MM-DD'), 500);
INSERT INTO Salary VALUES (2, 2002, 4500, 4500 * 0.04, 4500 + (4500 * 0.04),
TO_DATE('2024-07-31', 'YYYY-MM-DD'), NULL);
INSERT INTO Salary VALUES (3, 2003, 4800, 4800 * 0.06, 4800 + (4800 * 0.06) + 300,
TO_DATE('2024-07-31', 'YYYY-MM-DD'), 300);
INSERT INTO Salary VALUES (4, 2004, 4700, 4700 * 0.05, 4700 + (4700 * 0.05),
TO_DATE('2024-07-31', 'YYYY-MM-DD'), NULL);
INSERT INTO Salary VALUES (5, 2005, 4600, 4600 * 0.04, 4600 + (4600 * 0.04) + 250,
TO_DATE('2024-07-31', 'YYYY-MM-DD'), 250);
INSERT INTO Salary VALUES (6, 2006, 4800, 4800 * 0.06, 4800 + (4800 * 0.06),
TO_DATE('2024-07-31', 'YYYY-MM-DD'), NULL);
INSERT INTO Salary VALUES (7, 2007, 4500, 4500 * 0.05, 4500 + (4500 * 0.05),
TO_DATE('2024-07-31', 'YYYY-MM-DD'), NULL);
INSERT INTO Salary VALUES (8, 2008, 4200, 4200 * 0.04, 4200 + (4200 * 0.04) + 150,
TO_DATE('2024-07-31', 'YYYY-MM-DD'), 150);
INSERT INTO Salary VALUES (9, 2009, 4600, 4600 * 0.06, 4600 + (4600 * 0.06),
TO_DATE('2024-07-31', 'YYYY-MM-DD'), NULL);
INSERT INTO Salary VALUES (10, 2010, 4700, 4700 * 0.05, 4700 + (4700 * 0.05) + 200,
TO_DATE('2024-07-31', 'YYYY-MM-DD'), 200);
INSERT INTO Salary VALUES (11, 2011, 4600, 4600 * 0.04, 4600 + (4600 * 0.04),
TO_DATE('2024-07-31', 'YYYY-MM-DD'), NULL);
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INSERT INTO Salary VALUES (12, 2012, 4800, 4800 * 0.06, 4800 + (4800 * 0.06) + 100,
TO_DATE('2024-07-31', 'YYYY-MM-DD'), 100);
INSERT INTO Salary VALUES (13, 2013, 4500, 4500 * 0.05, 4500 + (4500 * 0.05),
TO_DATE('2024-07-31', 'YYYY-MM-DD'), NULL);
INSERT INTO Salary VALUES (14, 2014, 4200, 4200 * 0.04, 4200 + (4200 * 0.04) + 180,
TO_DATE('2024-07-31', 'YYYY-MM-DD'), 180);
INSERT INTO Salary VALUES (15, 2015, 4600, 4600 * 0.06, 4600 + (4600 * 0.06),
TO_DATE('2024-07-31', 'YYYY-MM-DD'), NULL);

--------------------------------------------------------------------------------
-- Insert data into Customer table
--------------------------------------------------------------------------------
-- Insert Customer records
INSERT INTO Customer VALUES (3001, 'Ali Bin Ahmad', '011-9988776', '1 Jalan Pudu',
'ali@customer.my', 2001);
INSERT INTO Customer VALUES (3002, 'Siti Nurhaliza', '012-4485667', '2 Jalan Imbi',
'siti@customer.my', 2001);
INSERT INTO Customer VALUES (3003, 'Tan Chee Keong', '013-3344556', '3 Jalan Alor',
'tan@customer.my', 2002);
INSERT INTO Customer VALUES (3004, 'Lim Pei Yen', '016-2230445', '4 Jalan Burma',
'lim@customer.my', 2002);
INSERT INTO Customer VALUES (3005, 'Cheong Wei Ling', '011-5566778', '5 Jalan
Macalister', 'cheong@customer.my', 2003);
INSERT INTO Customer VALUES (3006, 'Teoh Hock Chye', '012-6677889', '6 Jalan Anson',
'teoh@customer.my', 2003);
INSERT INTO Customer VALUES (3007, 'Ravi Shankar', '013-7788990', '7 Jalan Trus',
'ravi@customer.my', 2004);
INSERT INTO Customer VALUES (3008, 'Suresh Narayan', '016-8899001', '8 Jalan Wong
Ah Fook', 'suresh@customer.my', 2004);
INSERT INTO Customer VALUES (3009, 'Sally Smith', '011-9900112', '9 Jalan Ibrahim',
'Sally@customer.my', 2005);
INSERT INTO Customer VALUES (3010, 'Rosli Ahmad', '012-0011223', '10 Jalan Tun Fuad',
'rosli@customer.my', 2005);
INSERT INTO Customer VALUES (3011, 'Mariam Salleh', '013-1122334', '11 Jalan Pantai',
'mariam@customer.my', 2001);
INSERT INTO Customer VALUES (3012, 'Jonathan Lee', '016-2233445', '12 Jalan Lintas',
'jonathan@customer.my', 2002);
INSERT INTO Customer VALUES (3013, 'Josephine Lee', '011-3344556', '13 Jalan
Padungan', 'josephine@customer.my', 2003);
INSERT INTO Customer VALUES (3014, 'Mohd Fadzil', '012-4455667', '14 Jalan Pending',
'fadzil@customer.my', 2004);
INSERT INTO Customer VALUES (3015, 'Christina Lim', '013-5266778', '15 Jalan Tabuan',
'christina@customer.my', 2005);
INSERT INTO Customer VALUES (3016, 'Arif Bin Mohd', '016-6677889', '16 Jalan Satok',
'arif@customer.my', 2001);
```

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INSERT INTO Customer VALUES (3017, 'Ng Wei Liang', '011-7788990', '17 Jalan Song', 'ng@customer.my', 2002);
INSERT INTO Customer VALUES (3018, 'Lee Kah Wei', '012-8899001', '18 Jalan Kuching', 'lee@customer.my', 2003);
INSERT INTO Customer VALUES (3019, 'Chan Kok Leong', '013-9900112', '19 Jalan Kota Sentosa', 'chan@customer.my', 2004);
INSERT INTO Customer VALUES (3020, 'Azman Rahim', '016-0011223', '20 Jalan Rock', 'azman@customer.my', 2005);
INSERT INTO Customer VALUES (3021, 'John Doe', '011-1222333', '21 Jalan Ampang', 'john.doe@customer.my', 2001);
INSERT INTO Customer VALUES (3022, 'Jane Smith', '012-3344556', '22 Jalan Tun Razak', 'jane.smith@customer.my', 2002);
INSERT INTO Customer VALUES (3023, 'Alice Tan', '011-1222334', '23 Jalan Tun Ismail', 'alice@customer.my', 2001);
INSERT INTO Customer VALUES (3024, 'Bob Lim', '011-2233445', '24 Jalan Bukit Bintang', 'bob@customer.my', 2002);
INSERT INTO Customer VALUES (3025, 'Charlie Wong', '012-3144556', '25 Jalan Imbi', 'charlie@customer.my', 2003);
INSERT INTO Customer VALUES (3026, 'David Lee', '012-4055667', '26 Jalan Pudu', 'david@customer.my', 2004);
INSERT INTO Customer VALUES (3027, 'Eve Ong', '012-5566778', '27 Jalan Alor', 'eve@customer.my', 2005);



--------------------------------------------------------------------------------------------
-- Insert data into Package table
--------------------------------------------------------------------------------------------
INSERT INTO Package VALUES (1, 'KL City Tour', 'Explore the sights of Kuala Lumpur', 300, TO_DATE('2024-07-01', 'YYYY-MM-DD'), TO_DATE('2024-07-03', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (2, 'Penang Heritage Tour', 'Discover the heritage of Penang', 400, TO_DATE('2024-07-05', 'YYYY-MM-DD'), TO_DATE('2024-07-08', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (3, 'Johor Bahru Shopping Spree', 'Enjoy shopping in Johor Bahru', 250, TO_DATE('2024-07-10', 'YYYY-MM-DD'), TO_DATE('2024-07-11', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (4, 'Kota Kinabalu Adventure', 'Experience the adventures in Kota Kinabalu', 500, TO_DATE('2024-07-12', 'YYYY-MM-DD'), TO_DATE('2024-07-16', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (5, 'Kuching Cultural Trip', 'Immerse in the culture of Kuching', 350, TO_DATE('2024-07-17', 'YYYY-MM-DD'), TO_DATE('2024-07-19', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (6, 'Langkawi Island Getaway', 'Relax at Langkawi Island', 600, TO_DATE('2024-07-20', 'YYYY-MM-DD'), TO_DATE('2024-07-23', 'YYYY-MM-DD'));
```

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INSERT INTO Package VALUES (7, 'Genting Highlands Fun', 'Have fun at Genting Highlands', 200, TO_DATE('2024-07-24', 'YYYY-MM-DD'), TO_DATE('2024-07-25', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (8, 'Malacca Historical Tour', 'Learn the history of Malacca', 300, TO_DATE('2024-07-26', 'YYYY-MM-DD'), TO_DATE('2024-07-28', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (9, 'Cameron Highlands Retreat', 'Relax at Cameron Highlands', 400, TO_DATE('2024-07-29', 'YYYY-MM-DD'), TO_DATE('2024-08-01', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (10, 'Redang Island Adventure', 'Enjoy the beach at Redang Island', 500, TO_DATE('2024-08-02', 'YYYY-MM-DD'), TO_DATE('2024-08-06', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (11, 'KL and Penang Combo', 'Explore both Kuala Lumpur and Penang', 700, TO_DATE('2024-08-10', 'YYYY-MM-DD'), TO_DATE('2024-08-15', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (12, 'Cultural and Nature Tour', 'Experience the culture in Kuching and nature in Cameron Highlands', 750, TO_DATE('2024-08-17', 'YYYY-MM-DD'), TO_DATE('2024-08-22', 'YYYY-MM-DD'));
INSERT INTO Package VALUES (13, 'Beach and Historical Sites', 'Relax at Redang Island and learn history in Malacca', 800, TO_DATE('2024-08-25', 'YYYY-MM-DD'), TO_DATE('2024-08-30', 'YYYY-MM-DD'));
--------------------------------------------------------------------------------
-- Insert data into Reservation table
--------------------------------------------------------------------------------
-- Insert new synthetic data into Reservation table
INSERT INTO Reservation VALUES (101, 3001, 2001, TO_DATE('2024-07-15', 'YYYY-MM-DD'), TO_DATE('2024-07-20', 'YYYY-MM-DD'), TO_DATE('2024-07-25', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (102, 3001, 2001, TO_DATE('2024-07-25', 'YYYY-MM-DD'), TO_DATE('2024-07-30', 'YYYY-MM-DD'), TO_DATE('2024-08-05', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (103, 3002, 2002, TO_DATE('2024-07-15', 'YYYY-MM-DD'), TO_DATE('2024-07-20', 'YYYY-MM-DD'), TO_DATE('2024-07-25', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (104, 3002, 2002, TO_DATE('2024-08-01', 'YYYY-MM-DD'), TO_DATE('2024-08-05', 'YYYY-MM-DD'), TO_DATE('2024-08-10', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (105, 3002, 2002, TO_DATE('2024-08-10', 'YYYY-MM-DD'), TO_DATE('2024-08-15', 'YYYY-MM-DD'), TO_DATE('2024-08-20', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (106, 3003, 2002, TO_DATE('2024-06-10', 'YYYY-MM-DD'), TO_DATE('2024-07-10', 'YYYY-MM-DD'), TO_DATE('2024-07-11', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (107, 3004, 2002, TO_DATE('2024-06-15', 'YYYY-MM-DD'), TO_DATE('2024-07-12', 'YYYY-MM-DD'), TO_DATE('2024-07-16', 'YYYY-MM-DD'), 'Confirmed', NULL);
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INSERT INTO Reservation VALUES (108, 3005, 2003, TO_DATE('2024-06-20', 'YYYY-MM-DD'), TO_DATE('2024-07-17', 'YYYY-MM-DD'), TO_DATE('2024-07-19', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (109, 3006, 2003, TO_DATE('2024-06-25', 'YYYY-MM-DD'), TO_DATE('2024-07-20', 'YYYY-MM-DD'), TO_DATE('2024-07-23', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (110, 3007, 2004, TO_DATE('2024-06-30', 'YYYY-MM-DD'), TO_DATE('2024-07-24', 'YYYY-MM-DD'), TO_DATE('2024-07-25', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (111, 3008, 2004, TO_DATE('2024-07-01', 'YYYY-MM-DD'), TO_DATE('2024-07-26', 'YYYY-MM-DD'), TO_DATE('2024-07-28', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (112, 3009, 2005, TO_DATE('2024-07-05', 'YYYY-MM-DD'), TO_DATE('2024-07-29', 'YYYY-MM-DD'), TO_DATE('2024-08-01', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (113, 3010, 2005, TO_DATE('2024-07-10', 'YYYY-MM-DD'), TO_DATE('2024-08-02', 'YYYY-MM-DD'), TO_DATE('2024-08-06', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (114, 3011, 2001, TO_DATE('2024-07-15', 'YYYY-MM-DD'), TO_DATE('2024-07-01', 'YYYY-MM-DD'), TO_DATE('2024-07-03', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (115, 3012, 2001, TO_DATE('2024-07-20', 'YYYY-MM-DD'), TO_DATE('2024-07-05', 'YYYY-MM-DD'), TO_DATE('2024-07-08', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (116, 3013, 2002, TO_DATE('2024-07-25', 'YYYY-MM-DD'), TO_DATE('2024-07-10', 'YYYY-MM-DD'), TO_DATE('2024-07-11', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (117, 3014, 2002, TO_DATE('2024-07-30', 'YYYY-MM-DD'), TO_DATE('2024-07-12', 'YYYY-MM-DD'), TO_DATE('2024-07-16', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (118, 3015, 2003, TO_DATE('2024-08-01', 'YYYY-MM-DD'), TO_DATE('2024-07-17', 'YYYY-MM-DD'), TO_DATE('2024-07-19', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (119, 3016, 2003, TO_DATE('2024-08-05', 'YYYY-MM-DD'), TO_DATE('2024-07-20', 'YYYY-MM-DD'), TO_DATE('2024-07-23', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (120, 3017, 2004, TO_DATE('2024-08-10', 'YYYY-MM-DD'), TO_DATE('2024-07-24', 'YYYY-MM-DD'), TO_DATE('2024-07-25', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (121, 3018, 2004, TO_DATE('2024-08-15', 'YYYY-MM-DD'), TO_DATE('2024-07-26', 'YYYY-MM-DD'), TO_DATE('2024-07-28', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (122, 3019, 2005, TO_DATE('2024-08-20', 'YYYY-MM-DD'), TO_DATE('2024-07-29', 'YYYY-MM-DD'), TO_DATE('2024-08-01', 'YYYY-MM-DD'), 'Confirmed', NULL);
```

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INSERT INTO Reservation VALUES (123, 3020, 2005, TO_DATE('2024-08-25', 'YYYY-MM-DD'), TO_DATE('2024-08-02', 'YYYY-MM-DD'), TO_DATE('2024-08-06', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (124, 3021, 2001, TO_DATE('2024-09-01', 'YYYY-MM-DD'), TO_DATE('2024-09-10', 'YYYY-MM-DD'), TO_DATE('2024-09-20', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (125, 3022, 2002, TO_DATE('2024-09-15', 'YYYY-MM-DD'), TO_DATE('2024-09-25', 'YYYY-MM-DD'), TO_DATE('2024-10-05', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (128, 3023, 2001, TO_DATE('2024-11-01', 'YYYY-MM-DD'), TO_DATE('2024-11-10', 'YYYY-MM-DD'), TO_DATE('2024-11-15', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (129, 3023, 2001, TO_DATE('2024-11-05', 'YYYY-MM-DD'), TO_DATE('2024-11-15', 'YYYY-MM-DD'), TO_DATE('2024-11-20', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (130, 3024, 2002, TO_DATE('2024-11-10', 'YYYY-MM-DD'), TO_DATE('2024-11-20', 'YYYY-MM-DD'), TO_DATE('2024-11-25', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (131, 3024, 2002, TO_DATE('2024-11-15', 'YYYY-MM-DD'), TO_DATE('2024-11-25', 'YYYY-MM-DD'), TO_DATE('2024-11-30', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (132, 3025, 2003, TO_DATE('2024-11-20', 'YYYY-MM-DD'), TO_DATE('2024-11-30', 'YYYY-MM-DD'), TO_DATE('2024-12-05', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (133, 3025, 2003, TO_DATE('2024-11-25', 'YYYY-MM-DD'), TO_DATE('2024-12-05', 'YYYY-MM-DD'), TO_DATE('2024-12-10', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (134, 3026, 2004, TO_DATE('2024-11-30', 'YYYY-MM-DD'), TO_DATE('2024-12-10', 'YYYY-MM-DD'), TO_DATE('2024-12-15', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (135, 3026, 2004, TO_DATE('2024-12-05', 'YYYY-MM-DD'), TO_DATE('2024-12-15', 'YYYY-MM-DD'), TO_DATE('2024-12-20', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (136, 3027, 2005, TO_DATE('2024-12-10', 'YYYY-MM-DD'), TO_DATE('2024-12-20', 'YYYY-MM-DD'), TO_DATE('2024-12-25', 'YYYY-MM-DD'), 'Confirmed', NULL);
INSERT INTO Reservation VALUES (137, 3027, 2005, TO_DATE('2024-12-15', 'YYYY-MM-DD'), TO_DATE('2024-12-25', 'YYYY-MM-DD'), TO_DATE('2024-12-30', 'YYYY-MM-DD'), 'Confirmed', NULL);

--------------------------------------------------------------------------------
-- Insert data into Review table
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```
INSERT INTO Review VALUES (501, 3001, 101, 'Great tour of KL!', TO_DATE('2024-07-04', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (502, 3002, 102, 'Loved Penang!', TO_DATE('2024-07-09', 'YYYY-MM-DD'), 4);
INSERT INTO Review VALUES (503, 3002, 103, 'Shopping was amazing!', TO_DATE('2024-07-12', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (504, 3002, 104, 'Adventure of a lifetime!', TO_DATE('2024-07-17', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (505, 3003, 105, 'Cultural trip was enlightening.', TO_DATE('2024-07-20', 'YYYY-MM-DD'), 4);
INSERT INTO Review VALUES (506, 3003, 106, 'Island getaway was relaxing.', TO_DATE('2024-07-24', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (507, 3004, 107, 'Fun at Genting Highlands.', TO_DATE('2024-07-26', 'YYYY-MM-DD'), 4);
INSERT INTO Review VALUES (508, 3005, 108, 'Historical tour was informative.', TO_DATE('2024-07-29', 'YYYY-MM-DD'), 4);
INSERT INTO Review VALUES (509, 3006, 109, 'Relaxing retreat.', TO_DATE('2024-08-02', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (510, 3007, 110, 'Beach adventure was fantastic.', TO_DATE('2024-08-07', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (511, 3008, 111, 'KL City Tour was great.', TO_DATE('2024-07-04', 'YYYY-MM-DD'), 4);
INSERT INTO Review VALUES (512, 3009, 112, 'Penang Heritage Tour was excellent.', TO_DATE('2024-07-09', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (513, 3010, 113, 'Shopping in JB was fun.', TO_DATE('2024-07-12', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (514, 3011, 114, 'Kota Kinabalu Adventure was thrilling.', TO_DATE('2024-07-17', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (515, 3012, 115, 'Kuching Cultural Trip was insightful.', TO_DATE('2024-07-20', 'YYYY-MM-DD'), 4);
INSERT INTO Review VALUES (516, 3013, 116, 'Langkawi Island Getaway was perfect.', TO_DATE('2024-07-24', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (517, 3014, 117, 'Genting Highlands was exciting.', TO_DATE('2024-07-26', 'YYYY-MM-DD'), 4);
INSERT INTO Review VALUES (518, 3015, 118, 'Malacca Historical Tour was educational.', TO_DATE('2024-07-29', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (519, 3016, 119, 'Cameron Highlands was beautiful.', TO_DATE('2024-08-02', 'YYYY-MM-DD'), 5);
INSERT INTO Review VALUES (520, 3017, 120, 'Redang Island Adventure was unforgettable.', TO_DATE('2024-08-07', 'YYYY-MM-DD'), 5);

--------------------------------------------------------------------------------
-- Insert data into Payment table
--------------------------------------------------------------------------------
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```
INSERT INTO Payment VALUES (1, 101, 'Credit Card', TO_DATE('2024-06-01', 'YYYY-MM-DD'), 300 * 0.9, 'Paid');
INSERT INTO Payment VALUES (2, 102, 'Credit Card', TO_DATE('2024-06-05', 'YYYY-MM-DD'), 400 - 50, 'Paid');
INSERT INTO Payment VALUES (3, 103, 'Debit Card', TO_DATE('2024-06-10', 'YYYY-MM-DD'), 250, 'Paid');
INSERT INTO Payment VALUES (4, 104, 'Cash', TO_DATE('2024-06-15', 'YYYY-MM-DD'), 500 * 0.8, 'Paid');
INSERT INTO Payment VALUES (5, 105, 'Bank Transfer', TO_DATE('2024-06-20', 'YYYY-MM-DD'), 350, 'Paid');
INSERT INTO Payment VALUES (6, 106, 'Debit Card', TO_DATE('2024-06-10', 'YYYY-MM-DD'), 250, 'Paid');
INSERT INTO Payment VALUES (7, 107, 'Cash', TO_DATE('2024-06-15', 'YYYY-MM-DD'), 500 * 0.8, 'Paid');
INSERT INTO Payment VALUES (8, 108, 'Bank Transfer', TO_DATE('2024-06-20', 'YYYY-MM-DD'), 350, 'Paid');
INSERT INTO Payment VALUES (9, 109, 'Credit Card', TO_DATE('2024-06-25', 'YYYY-MM-DD'), 600 * 0.85, 'Paid');
INSERT INTO Payment VALUES (10, 110, 'Debit Card', TO_DATE('2024-06-30', 'YYYY-MM-DD'), 200, 'Paid');
INSERT INTO Payment VALUES (11, 111, 'Cash', TO_DATE('2024-07-01', 'YYYY-MM-DD'), 300, 'Paid');
INSERT INTO Payment VALUES (12, 112, 'Bank Transfer', TO_DATE('2024-07-05', 'YYYY-MM-DD'), 400 - 50, 'Paid');
INSERT INTO Payment VALUES (13, 113, 'Credit Card', TO_DATE('2024-07-10', 'YYYY-MM-DD'), 500, 'Paid');
INSERT INTO Payment VALUES (14, 114, 'Debit Card', TO_DATE('2024-07-15', 'YYYY-MM-DD'), 300 * 0.9, 'Paid');
INSERT INTO Payment VALUES (15, 115, 'Cash', TO_DATE('2024-07-20', 'YYYY-MM-DD'), 400 - 50, 'Paid');
INSERT INTO Payment VALUES (16, 116, 'Bank Transfer', TO_DATE('2024-07-25', 'YYYY-MM-DD'), 250, 'Paid');
INSERT INTO Payment VALUES (17, 117, 'Credit Card', TO_DATE('2024-07-30', 'YYYY-MM-DD'), 500 * 0.8, 'Paid');
INSERT INTO Payment VALUES (18, 118, 'Debit Card', TO_DATE('2024-08-01', 'YYYY-MM-DD'), 350, 'Paid');
INSERT INTO Payment VALUES (19, 119, 'Cash', TO_DATE('2024-08-05', 'YYYY-MM-DD'), 600 * 0.85, 'Paid');
INSERT INTO Payment VALUES (20, 120, 'Bank Transfer', TO_DATE('2024-08-10', 'YYYY-MM-DD'), 200, 'Paid');
INSERT INTO Payment VALUES (21, 121, 'Credit Card', TO_DATE('2024-08-15', 'YYYY-MM-DD'), 300, 'Paid');
INSERT INTO Payment VALUES (22, 122, 'Debit Card', TO_DATE('2024-08-20', 'YYYY-MM-DD'), 400, 'Paid');
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INSERT INTO Payment VALUES (23, 123, 'Cash', TO_DATE('2024-08-25', 'YYYY-MM-DD'),
500 * 0.8, 'Paid');

--------------------------------------------------------------------------------
-- Insert data into Invoice table
--------------------------------------------------------------------------------
INSERT INTO Invoice VALUES (1, 1, TO_DATE('2024-06-01', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (2, 2, TO_DATE('2024-06-05', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (3, 3, TO_DATE('2024-06-10', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (4, 4, TO_DATE('2024-06-15', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (5, 5, TO_DATE('2024-06-20', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (6, 6, TO_DATE('2024-06-10', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (7, 7, TO_DATE('2024-06-15', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (8, 8, TO_DATE('2024-06-20', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (9, 9, TO_DATE('2024-06-25', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (10, 10, TO_DATE('2024-06-30', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (11, 11, TO_DATE('2024-07-01', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (12, 12, TO_DATE('2024-07-05', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (13, 13, TO_DATE('2024-07-10', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (14, 14, TO_DATE('2024-07-15', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (15, 15, TO_DATE('2024-07-20', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (16, 16, TO_DATE('2024-07-25', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (17, 17, TO_DATE('2024-07-30', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (18, 18, TO_DATE('2024-08-01', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (19, 19, TO_DATE('2024-08-05', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (20, 20, TO_DATE('2024-08-10', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (21, 21, TO_DATE('2024-08-15', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (22, 22, TO_DATE('2024-08-20', 'YYYY-MM-DD'));
INSERT INTO Invoice VALUES (23, 23, TO_DATE('2024-08-25', 'YYYY-MM-DD'));

--------------------------------------------------------------------------------
-- Insert data into Destination table
--------------------------------------------------------------------------------
INSERT INTO Destination VALUES (1, 'Petronas Towers', 'Iconic skyscrapers in Kuala
Lumpur', 5, 'Landmark', 1);
INSERT INTO Destination VALUES (2, 'Penang Hill', 'Scenic hill in Penang', 10, 'Nature', 2);
INSERT INTO Destination VALUES (3, 'Legoland', 'Theme park in Johor Bahru', 15,
'Amusement Park', 3);
INSERT INTO Destination VALUES (4, 'Mount Kinabalu', 'Highest peak in Southeast Asia',
30, 'Mountain', 4);
INSERT INTO Destination VALUES (5, 'Sarawak Cultural Village', 'Living museum in
Kuching', 20, 'Cultural', 5);
INSERT INTO Destination VALUES (6, 'Langkawi Sky Bridge', 'Pedestrian bridge in
Langkawi', 25, 'Attraction', 6);
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INSERT INTO Destination VALUES (7, 'Genting Highlands', 'Hill resort in Genting', 35,
'Resort', 7);
INSERT INTO Destination VALUES (8, 'A Famosa', 'Historical fort in Malacca', 5, 'Historical',
8);
INSERT INTO Destination VALUES (9, 'BOH Tea Plantation', 'Tea plantation in Cameron
Highlands', 10, 'Nature', 9);
INSERT INTO Destination VALUES (10, 'Redang Island', 'Beautiful island in Terengganu', 40,
'Beach', 10);

--------------------------------------------------------------------------------
-- Insert data into Promotion table
--------------------------------------------------------------------------------
INSERT INTO Promotion VALUES (1, 'Discount', 'Summer Sale', TO_DATE('2024-06-01',
'YYYY-MM-DD'), TO_DATE('2024-07-31', 'YYYY-MM-DD'), '10% off on all packages', 10, 1);
INSERT INTO Promotion VALUES (2, 'Coupon', 'Early Bird', TO_DATE('2024-06-01', 'YYYY-
MM-DD'), TO_DATE('2024-07-31', 'YYYY-MM-DD'), 'RM50 off for early bookings', 50, 2);
INSERT INTO Promotion VALUES (3, 'Bundle', 'Family Package', TO_DATE('2024-06-01',
'YYYY-MM-DD'), TO_DATE('2024-07-31', 'YYYY-MM-DD'), '20% off for family packages', 20,
4);
INSERT INTO Promotion VALUES (4, 'Discount', 'Student Discount', TO_DATE('2024-06-01',
'YYYY-MM-DD'), TO_DATE('2024-07-31', 'YYYY-MM-DD'), '15% off for students', 15, 6);
INSERT INTO Promotion VALUES (5, 'Coupon', 'Weekend Special', TO_DATE('2024-06-01',
'YYYY-MM-DD'), TO_DATE('2024-07-31', 'YYYY-MM-DD'), 'RM30 off for weekend trips', 30,
8);

--------------------------------------------------------------------------------
-- Insert data into Transportation table
--------------------------------------------------------------------------------
INSERT INTO Transportation VALUES (1, 'Flight', 'Malaysia Airlines', TO_DATE('2024-07-
01', 'YYYY-MM-DD'), TO_DATE('2024-07-01', 'YYYY-MM-DD'), 1);
INSERT INTO Transportation VALUES (2, 'Bus', 'Rapid Penang', TO_DATE('2024-07-05',
'YYYY-MM-DD'), TO_DATE('2024-07-05', 'YYYY-MM-DD'), 2);
INSERT INTO Transportation VALUES (3, 'Car', 'Hertz', TO_DATE('2024-07-10', 'YYYY-MM-
DD'), TO_DATE('2024-07-10', 'YYYY-MM-DD'), 3);
INSERT INTO Transportation VALUES (4, 'Train', 'KTM', TO_DATE('2024-07-12', 'YYYY-
MM-DD'), TO_DATE('2024-07-12', 'YYYY-MM-DD'), 4);
INSERT INTO Transportation VALUES (5, 'Ferry', 'Langkawi Ferry', TO_DATE('2024-07-20',
'YYYY-MM-DD'), TO_DATE('2024-07-20', 'YYYY-MM-DD'), 6);

--------------------------------------------------------------------------------
-- Insert data into Accommodation table
--------------------------------------------------------------------------------
INSERT INTO Accommodation VALUES (1, 'Hotel', 'Hilton KL', 'Jalan Stesen Sentral', 1);
INSERT INTO Accommodation VALUES (2, 'Resort', 'Hard Rock Penang', 'Batu Ferringhi',
2);
```

```
INSERT INTO Accommodation VALUES (3, 'Hotel', 'DoubleTree JB', 'Jalan Ngee Heng', 3);
INSERT INTO Accommodation VALUES (4, 'Lodge', 'Kinabalu Park Lodge', 'Jalan Kinabalu',
4);
INSERT INTO Accommodation VALUES (5, 'Hotel', 'Pullman Kuching', 'Jalan Mathies', 5);
INSERT INTO Accommodation VALUES (6, 'Resort', 'Berjaya Langkawi', 'Burau Bay', 6);
INSERT INTO Accommodation VALUES (7, 'Hotel', 'Genting Grand', 'Genting Highlands', 7);
INSERT INTO Accommodation VALUES (8, 'Hotel', 'Casa del Rio', 'Jalan Kota Laksamana',
8);
INSERT INTO Accommodation VALUES (9, 'Hotel', 'Cameron Highlands Resort', 'Tanah
Rata', 9);
INSERT INTO Accommodation VALUES (10, 'Resort', 'The Taaras Beach & Spa Resort',
'Pulau Redang', 10);

--------------------------------------------------------------------------------
-- Insert data into Package_Destination table
--------------------------------------------------------------------------------

INSERT INTO Package_Destination VALUES (1, 1, 1, 100);
INSERT INTO Package_Destination VALUES (2, 2, 1, 150);
INSERT INTO Package_Destination VALUES (3, 3, 1, 120);
INSERT INTO Package_Destination VALUES (4, 4, 1, 200);
INSERT INTO Package_Destination VALUES (5, 5, 1, 180);
INSERT INTO Package_Destination VALUES (6, 6, 1, 220);
INSERT INTO Package_Destination VALUES (7, 7, 1, 140);
INSERT INTO Package_Destination VALUES (8, 8, 1, 160);
INSERT INTO Package_Destination VALUES (9, 9, 1, 170);
INSERT INTO Package_Destination VALUES (10, 10, 1, 200);
INSERT INTO Package_Destination VALUES (11, 1, 1, 110);
INSERT INTO Package_Destination VALUES (11, 2, 1, 150);
INSERT INTO Package_Destination VALUES (12, 5, 1, 180);
INSERT INTO Package_Destination VALUES (12, 9, 1, 170);
INSERT INTO Package_Destination VALUES (13, 10, 1, 200);
INSERT INTO Package_Destination VALUES (13, 8, 1, 160);

--------------------------------------------------------------------------------
-- Insert data into Package_Reservation table
--------------------------------------------------------------------------------

INSERT INTO Package_Reservation VALUES (1, 101, 1, 300);
INSERT INTO Package_Reservation VALUES (2, 102, 1, 400);
INSERT INTO Package_Reservation VALUES (1, 103, 1, 300);
INSERT INTO Package_Reservation VALUES (3, 104, 1, 250);
INSERT INTO Package_Reservation VALUES (4, 105, 1, 500);
INSERT INTO Package_Reservation VALUES (3, 106, 1, 250);
INSERT INTO Package_Reservation VALUES (4, 107, 1, 500);
```

```
INSERT INTO Package_Reservation VALUES (5, 108, 1, 350);
INSERT INTO Package_Reservation VALUES (6, 109, 1, 600);
INSERT INTO Package_Reservation VALUES (7, 110, 1, 200);
INSERT INTO Package_Reservation VALUES (8, 111, 1, 300);
INSERT INTO Package_Reservation VALUES (9, 112, 1, 400);
INSERT INTO Package_Reservation VALUES (10, 113, 1, 500);
INSERT INTO Package_Reservation VALUES (1, 114, 1, 300);
INSERT INTO Package_Reservation VALUES (2, 115, 1, 400);
INSERT INTO Package_Reservation VALUES (3, 116, 1, 250);
INSERT INTO Package_Reservation VALUES (4, 117, 1, 500);
INSERT INTO Package_Reservation VALUES (5, 118, 1, 350);
INSERT INTO Package_Reservation VALUES (6, 119, 1, 600);
INSERT INTO Package_Reservation VALUES (7, 120, 1, 200);
INSERT INTO Package_Reservation VALUES (8, 121, 1, 300);
INSERT INTO Package_Reservation VALUES (9, 122, 1, 400);
INSERT INTO Package_Reservation VALUES (10, 123, 1, 500);
INSERT INTO Package_Reservation VALUES (1, 124, 1, 300);
INSERT INTO Package_Reservation VALUES (2, 124, 1, 400);
INSERT INTO Package_Reservation VALUES (3, 124, 1, 500);
INSERT INTO Package_Reservation VALUES (4, 125, 1, 250);
INSERT INTO Package_Reservation VALUES (5, 125, 1, 350);
INSERT INTO Package_Reservation VALUES (6, 125, 1, 450);
INSERT INTO Package_Reservation VALUES (11, 128, 1, 700);
INSERT INTO Package_Reservation VALUES (12, 129, 1, 750);
INSERT INTO Package_Reservation VALUES (13, 130, 1, 800);
INSERT INTO Package_Reservation VALUES (11, 131, 1, 700);
INSERT INTO Package_Reservation VALUES (12, 132, 1, 750);
INSERT INTO Package_Reservation VALUES (13, 133, 1, 800);
INSERT INTO Package_Reservation VALUES (11, 134, 1, 700);
INSERT INTO Package_Reservation VALUES (12, 135, 1, 750);
INSERT INTO Package_Reservation VALUES (13, 136, 1, 800);
INSERT INTO Package_Reservation VALUES (11, 137, 1, 700);
```

`Part 4: Query a database (each sub-part carries 4 marks; max of 20 marks)

No	Query Description
a	The travel agency management wants to know which customer has made the most reservations in 2024 and get the details of their reservations. Use the EXTRACT function to ensure the data is from the current year.

```sql
SELECT CustomerID, CusName
FROM Customer
WHERE CustomerID IN (
    SELECT CustomerID
    FROM Reservation
    WHERE EXTRACT(YEAR FROM ReserveDate) = 2024
    GROUP BY CustomerID
    HAVING COUNT(*) = (
        SELECT MAX(Cnt)
        FROM (SELECT CustomerID, COUNT(*) AS Cnt
              FROM Reservation
              WHERE EXTRACT(YEAR FROM ReserveDate) = 2024
              GROUP BY CustomerID)
    )
);
```

Results | Explain | Describe | Saved SQL | History

CUSTOMERID	CUSNAME
3002	Siti Nurhaliza

1 rows returned in 0.03 seconds Download

b	The marketing department of Travoka Travel Agency wants to identify high-spending customers who frequently book packages. They are interested in analyzing the total amount spent by each customer on different packages, especially focusing on those customers who have made payments exceeding RM500 for any package. This information will be used to tailor special promotions and personalized offers to these valued customers.

```
1  SELECT c.CustomerID,c.CusName,p.PackageName,SUM(py.Amount) AS TotalSpent
2  FROM Customer c
3  JOIN Reservation r ON c.CustomerID = r.CustomerID
4  JOIN Payment py ON r.ReservationID = py.ReservationID
5  JOIN Package_Reservation pr ON r.ReservationID = pr.ReservationID
6  JOIN Package p ON pr.PackageID = p.PackageID
7  WHERE py.Amount > 500
8  GROUP BY c.CustomerID, c.CusName, p.PackageName
9  HAVING SUM(py.Amount) > 500;
```

Results | Explain | Describe | Saved SQL | History

CUSTOMERID	CUSNAME	PACKAGENAME	TOTALSPENT
3006	Teoh Hock Chye	Langkawi Island Getaway	600
3016	Arif Bin Mohd	Langkawi Island Getaway	600

2 rows returned in 0.05 seconds Download

c	The HR department wants to find out the agents whose names start with the letter 'S' and who have a supervisor assigned to them. They also want to know the name of their supervisor.

```sql
SELECT a1.AgentName AS Agent, a2.AgentName AS Supervisor
FROM Agent a1
LEFT JOIN Agent a2 ON a1.SupervisorID = a2.AgentID
WHERE a1.AgentName LIKE 'S%' AND a2.AgentName IS NOT NULL;
```

Results | Explain | Describe | Saved SQL | History

AGENT	SUPERVISOR
Siti Aminah	Ahmad Ali
Suresh Kumar	Ravi Subramaniam

2 rows returned in 0.00 seconds Download

d	Travoka's management wants to identify customers who have registered but have never made a reservation. This information is important for the marketing team to target these customers with special promotions to encourage them to book their first trip.

```sql
SELECT CustomerID, CusName, CusContact, CusEmail
FROM Customer
WHERE CustomerID IN (
    SELECT CustomerID
    FROM Reservation
    MINUS
    SELECT CustomerID
    FROM Review
);
```

Results | Explain | Describe | Saved SQL | History

CUSTOMERID	CUSNAME	CUSCONTACT	CUSEMAIL
3018	Lee Kah Wei	0128899001	lee@customer.my
3019	Chan Kok Leong	0139900112	chan@customer.my
3020	Azman Rahim	0160011223	azman@customer.my
3021	John Doe	0111222333	john.doe@customer.my
3022	Jane Smith	0123344556	jane.smith@customer.my
3023	Alice Tan	0111222334	alice@customer.my
3024	Bob Lim	0112233445	bob@customer.my
3025	Charlie Wong	0123144556	charlie@customer.my
3026	David Lee	0124055667	david@customer.my
3027	Eve Ong	0125566778	eve@customer.my

10 rows returned in 0.04 seconds Download

e	Calculate the total amount paid by customer Sally Smith for her most current booking.

```sql
SELECT SUM(p.amount) AS TotalAmountPaid
From Payment p
JOIN Reservation r on p.ReservationID = r.ReservationID
JOIN Customer c on r.customerID = c.CustomerID
Where c.cusname = 'Sally Smith'
AND r.ReserveDate = (
    SELECT MAX(RESERVEDATE)
    FROM Reservation
    Where CustomerID = c.CustomerID
)
```

Results | Explain | Describe | Saved SQL | History

TOTALAMOUNTPAID
400

1 rows returned in 0.06 seconds Download

Part 5: Teamwork and Presentation (20 marks)

No	ID	Name	Questions to present
i)	22067128	LIANG YRENN DA	Explain and run part 4 Question c Run the DBscript
ii)	22039069	OOI JIN YON	Explain and run part 4 Question d Explain and run part 4 Question e
iii)	22110597	YASHINI A/P CHANDRAN	Explain and run part 4 Question a Explain various key in each table
iv)	21056395	MUHAMMAD IKRAM BIN MOHAMED IQBAL	Explain and run part 4 Question b Explain index and constraint